More generally, if a given system has a first integral (conserved quantity) $I = I(z)$, different from the energy, which is preserved along trajectories, then the average of this conserved quantity (and any other quantity that depends upon it) will be dependent on the choice of initial condition.

5.4.1 The Concept of Ergodicity

While being mindful of the examples given above, there are many systems which, at least for a useful range of energy, appear to have the property that averages are independent of initial condition. Systems for which the trajectory averages are the same as phase space averages on the energy surface are said to be *ergodic*.[6] The concept of ergodicity can be introduced in the setting of general dynamical systems, but it is very challenging to analyze this property for realistic systems. An analogous concept for stochastic differential equations will be introduced in the next chapter; in this setting of randomized models ergodicity is much more amenable to mathematical study. Nonetheless, we motivate the concept in this section using deterministic systems and numerical examples.

The formal definition of ergodicity takes one of several forms. For our current purposes, we opt for a definition that is close to our intended goal (the computation of averages).

Definition 5.1 A Hamiltonian system is said to be ergodic on an energy surface Σ_{E} (microcanonically ergodic) if, for almost all trajectories Γ emanating from initial conditions on Σ_{E}, and for any observable g, the following holds

$$\bar{g}_{\Gamma} = \mathrm{Av}_{H=\mathrm{E}}(g),$$

where $\bar{g}_{\Gamma}$ is defined in (5.9). By "almost all" we mean that there could be a low-dimensional subset for which the trajectory averages fail to equal the space averages. The term "almost all" thus allows for the existence of some fixed points and periodic orbits, which are almost certain to be present in a deterministic system.

Alternatively, we may say that a dynamical system as ergodic if the probability measure of any invariant set $\mathcal{A}$ of the flow is either 0 or 1. An invariant set $\mathcal{A}$ of the flow is one such that

$$\mathscr{F}_t(\mathcal{A}) = \mathcal{A},$$

for all $t \neq 0$. To see that this is the same concept of ergodicity as defined above, let χ be the "indicator function" of the complement of $\mathcal{A}$ whose values is zero if $z \in \mathcal{A}$

[6]The term "ergodic" was coined by Boltzmann in 1887. He joined the Greek term *ergon* ("action," "work") with *hodos* ("path") to refer to an energy-path. An ergodic path one that traverses the entire energy surface eventually touching all points thereof.

and one if $z \in \Sigma_E \setminus \mathcal{A}$ (the symbol $\setminus$ means the removal of the second set from the first). Then for any trajectory starting in $\mathcal{A}$, we have that the average of χ along Γ vanishes, i.e.

$$\bar{\chi}_\Gamma = \lim_{\tau\to\infty} \tau^{-1} \int_0^\tau \chi(z(t))\mathrm{d}t = 0,$$

since $\mathcal{A}$ is an invariant set.

On the other hand, the average of χ on Σ_E is just the probability measure of $\Sigma_E \setminus \mathcal{A}$. If we assume the system is ergodic, we must have $\Pr(\Sigma_E \setminus \mathcal{A}) = 0$, thus $\Pr(\mathcal{A}) = 1$. Since, according to the definition, equality of time and space averages is allowed to fail only on a set of probability measure zero, we have that either $\Pr(\mathcal{A}) = 0$ or $\Pr(\mathcal{A}) = 1$.

If a system is not ergodic, then there will be an invariant set $\mathcal{A}$ whose probability measure is strictly between zero and one. The trajectories emanating from this set are unable to explore the entire energy surface and the corresponding trajectory averages will, at least for some observables, be different than the spatial average of that observable. A system which is not microcanonically ergodic has an energy surface that can be partitioned into subdomains of nonzero measure in which trajectories are forever trapped.

Often, the strict definition of ergodicity must be relaxed to accommodate the realistic modelling setting; for example only some portion of the energy surface may be accessible by trajectories in the time-interval available, due to the cost of computing steps with a numerical method. For a more detailed discussion of these and other ergodicity issues in the context of deterministic molecular dynamics, see the articles of Tupper [379, 381].

Given a system with energy of the form $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$, what can be said about the ergodicity of the system on the energy surface $H = \mathrm{E}$? Coverage of the set may follow in certain cases from periodic or quasiperiodic motion in which the trajectories wrap around the surface filling in space.

As a very simple example, the harmonic oscillator is always ergodic on its energy surfaces (i.e. energy curves, in this case). Some examples of two-dimensional systems are shown in Fig. 5.3.

Ergodicity is a concept that often becomes more relevant in systems with many degrees of freedom. For example if one thinks of having not just one but many double well oscillators connected together in some way so that they can exchange energy, then it is likely that the energy can accumulate, given enough time, in one of the oscillators, enabling it to overcome the barrier.

Example 5.4 (2D Harmonic System) Consider a two-dimensional harmonic system, i.e. the system with Hamiltonian

$$H = p_1^2/2 + p_2^2/2 + \Omega_1^2 q_1^2/2 + \Omega_2^2 q_2^2/2,$$